

Revisiting the Notion of Representability

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Our current notion of quasi-representability of relations involves perturbations associated to each of our representable relations. This leads to two complications with our semantics:

1. The projection associated to $F(cc')$ is not simply the composition of the projections for Fc and Fc' (unlike the original term semantics); it instead involves an additional perturbation. The additional perturbation is inserted in the relational semantics as part of the compositional construction of quasi-right-representability of $F(cc')$ given quasi-left-representability of c and c' . For a specific example where this perturbation results in observably different behavior compared to the original term semantics, see the Appendix of this document.
2. The (quasi-)order equivalence of two quasi-representable relations does not follow from sharing the same embedding and projection morphisms. Instead, we needed to define a separate primitive notion of quasi-equivalence where we have squares with perturbations on the side.

There is an alternative, more principled approach to perturbations and quasi-representable relations. The idea is to define a double category such that what were *quasi-representable* relations in the original double category of value objects/morphisms/relations/squares are now actually *representable* relations in the usual sense (i.e. with identity morphisms rather than perturbations).

There are two key differences between this formulation of perturbations and the original. First, in the new double setup, the vertical morphisms come equipped with an *action* on perturbations. Second, the squares in the new double category allow for an explicit synchronization of the vertical morphisms by a perturbation on both sides. We elaborate on these differences below, but first, we must make an important change to the definition of value and computation types, requiring that the monoid of syntactic perturbations be commutative. We will see why this is needed later, but for now note the new definition of value/computation objects.

Definition 0.1. A *value object* consists of a predomain A , a **commutative** monoid of perturbations M_A , and an homomorphism of monoids $i_A : M_A \rightarrow \text{Endo}_{\approx}(A)$.

A *computation object* is defined similarly: an error domain B , a **commutative monoid** M_B and a homomorphism of monoids i_B .

Note that $\text{Endo}_{\approx}(A)$ is not a commutative monoid, so we will need a more complicated universal property to define a homomorphism from a freely-constructed commutative monoid into $\text{Endo}_{\approx}(A)$. We will discuss this further below.

0.1 Vertical Morphisms

The vertical morphisms are defined as follows:

Definition 0.2. Given value objects $A = (|A|, M_A, i_A)$ and $A' = (|A'|, M_{A'}, i_{A'})$, a *perturbation-preserving morphism* consists of

- A morphism of the underlying predomains $f : |A| \rightarrow |A'|$
- Two homomorphisms of monoids $h^{\rightarrow} : M_A \rightarrow M_{A'}$ and $h^{\leftarrow} : M_{A'} \rightarrow M_A$

- Two coherence conditions: for all $m \in M_A$, we have $f \circ i_A(m) = i_{A'}(h^\rightarrow(m)) \circ f$:

$$\begin{array}{ccc} A & \xrightarrow{f} & A' \\ i_A(m) \downarrow & & \downarrow i_{A'}(h^\rightarrow(m)) \\ A & \xrightarrow{f} & A' \end{array}$$

and likewise for all $m' \in M_{A'}$, we have $f \circ i_A(h^\leftarrow(m')) = i_{A'}(m') \circ f$:

$$\begin{array}{ccc} A & \xrightarrow{f} & A' \\ i_A(h^\leftarrow(m')) \downarrow & & \downarrow i_{A'}(m') \\ A & \xrightarrow{f} & A' \end{array}$$

We make the analogous definition for computation objects.

Remark 1. Not all predomain/error domain morphisms extend to an action on perturbations in this way. The key counterexample is the arrow type constructor. The issue is that the perturbations of $A \rightarrow B$ are defined as $M_A^{op} \oplus M_B$, i.e., both M_A and M_B are in covariant positions. But the universal property for the arrow type says that

$$\mathcal{V}(\Gamma, U(A \rightarrow B)) \cong \mathcal{V}(\Gamma \times A, UB)$$

On the left, the perturbations are monoid homomorphisms $M_\Gamma \multimap (M_A^{op} \oplus M_B)$ whereas on the right they are monoid homomorphisms $(M_\Gamma \oplus M_A) \multimap M_B$.

0.2 Squares

A square in this double category will be defined as a square in the usual sense, but where we are allowed to synchronize the vertical morphisms on either side with a perturbation. More concretely:

Definition 0.3. Let $f : A_i \rightarrow A_o$ and $g : A'_i \rightarrow A'_o$ be perturbation-preserving predomain morphisms and let $c_i : A_i \multimap A'_i$ and $c_o : A_o \multimap A'_o$ be predomain relations. We say that there is a **perturbed square** between f and g , written $f \sqsubseteq_{\text{ptb}_{c_o}^{c_i}} g$, if there *merely exist* perturbations m_f and m_g such that $i(m_f) \circ f \sqsubseteq_{c_o}^{c_i} i(m_g) \circ g$. We make the analogous definition for error domains.

Perturbed squares can be composed vertically and horizontally, as shown in the following two lemmas. See the Appendix for proofs.

Lemma 0.1. Suppose $f_1 \sqsubseteq_{\text{ptb}_{c_2}^{c_1}} g_1$ and $f_2 \sqsubseteq_{\text{ptb}_{c_3}^{c_2}} g_2$. Then $(f_2 \circ f_1) \sqsubseteq_{\text{ptb}_{c_3}^{c_1}} (g_2 \circ g_1)$.

Lemma 0.2. If $f \sqsubseteq_{\text{ptb}_{c_o}^{c_i}} g$ and $g \sqsubseteq_{\text{ptb}_{c'_o}^{c'_i}} h$, then $f \sqsubseteq_{\text{ptb}_{c_o \odot c'_o}^{c_i \odot c'_i}} h$.

The important thing to note here is that the horizontal composition uses the fact that the monoids are commutative, as well as the push-pull property of the relations. For details, see the proof in the appendix.

0.3 Horizontal Morphisms

0.3.1 Push-Pull

As we have seen in the previous section, the push-pull property is necessary to allow us to compose squares horizontally. But as it turns out, our new definition of vertical morphisms allows us to *derive* push-pull for relations that happen to be the restriction of an identity relation along a morphism.

Definition 0.4. Let $f : A \rightarrow A'$ be a perturbation-preserving morphism. The **(left-)restriction** of $r(A')$ along f , denoted $f^*r(A') : A \multimap A'$, is defined by $x f^*r(A') y := f x \sqsubseteq_{A'} y$.

Likewise let $g : A' \rightarrow A$ be a perturbation-preserving morphism. The **(right-)restriction** of $r(A)$ along g , denoted $r(A)^*g : A \multimap A'$, is defined by $x r(A)^*g y := x \sqsubseteq_A g y$.

Lemma 0.3. *Let $f : A \rightarrow A'$ be a perturbation-preserving morphism. Then the relation $f^*r(A')$ has the push-pull property.*

*Similarly, let $g : A' \rightarrow A$ be a perturbation-preserving morphism. Then the relation $r(A)^*g$ has the push-pull property.*

0.3.2 Representability

Recall that the notion of *quasi*-representability of a relation involves a perturbation corresponding to the morphism opposite the cast. For each of the UPL, UPR, DNL, and DNR rules there is a distinguished perturbation that appears in the relevant square. In the double category of casts we have just defined, the perturbations are moved into the definition of the squares themselves. Thus, we can define representability of a relation as the familiar notion of representability of a horizontal morphism in a double category. In particular:

Definition 0.5. Let A and A' be value objects, and let $c : A \multimap A'$ be a predomain relation. We say that c is *left-representable* by a perturbation-preserving morphism $e_c : A \rightarrow A'$ if there are squares $\text{UPL} : e_c \sqsubseteq \text{ptb}_{r(A')}^c \text{id}_{A'}$ and $\text{UPR} : \text{id}_A \sqsubseteq \text{ptb}_c^{r(A)} e_c$. We say that c is *left-representable* if it is left-representable by e_c for some e_c .

We say that c is *right-representable* by a perturbation-preserving morphism $p_c : A' \rightarrow A$ if there are squares $\text{DNR} : \text{id}_A \sqsubseteq \text{ptb}_{r(A)}^c p_c$ and $\text{DNL} : p_c \sqsubseteq \text{ptb}_c^{r(A')} \text{id}_{A'}$.

We define the notion of representability for error domain relations analogously.

An important consequence of this definition is that if c is left (resp. right) representable, then it is actually determined *up to perturbation* by the embedding (resp. projection) morphism. More specifically, suppose c is left-representable by a morphism e_c . Recall the relation $e_c^*r(A')$, i.e., the left restriction of the reflexive relation on A' along the morphism e_c . The fact that c is left-representable by e_c implies that c is *order-equivalent* to $e_c^*r(A')$, as shown in the lemma below.

Lemma 0.4. *Let $c : A \multimap A'$ be a predomain relation satisfying push-pull. If c is left-representable by e_c , then there is a perturbed square $\text{id} \sqsubseteq \text{ptb}_{e_c^*r(A')}^c \text{id}$ and a perturbed square $\text{id} \sqsubseteq \text{ptb}_c^{e_c^*r(A')} \text{id}$.*

The analogous result is true for any error domain relation $d : B \multimap B'$ satisfying push-pull.

Similarly, right-representability by a morphism p_c implies order equivalence with the right restriction by p_c :

Lemma 0.5. *Let $c : A \multimap A'$ be a predomain relation satisfying push-pull. If c is right-representable by p_c , then there is a perturbed square $\text{id} \sqsubseteq \text{ptb}_{r(A)^*p_c}^c \text{id}$ and a perturbed square $\text{id} \sqsubseteq \text{ptb}_c^{r(A)^*p_c} \text{id}$.*

The analogous result is true for any error domain relation $d : B \multimap B'$ satisfying push-pull.

The perturbations witnessing the order equivalence are hidden in the squares; from the viewpoint of the double category, the two relations are order equivalent.

Remark 2. It is *not* necessarily the case that if a relation $c : A \multimap A'$ is left-representable by e_c , then it is actually *equal* to the restriction $e_c^*r(A')$. Nor is a right-representable relation necessarily equal to a restriction. For a simple example, consider the relation $F\text{Inj}_{\rightarrow} : FU(D \rightarrow FD) \multimap FD$. This is quasi-right-representable in the old semantics, and hence it is right-representable in the new semantics where the perturbation is now part of the square. But the right-restriction of the reflexive relation $FU(D \rightarrow FD)$ along the projection morphism p is not the same relation as $F\text{Inj}_{\rightarrow}$. Indeed, the restriction relates $x : FU(D \rightarrow FD)$ to $y : FD$ if and only if $x \sqsubseteq_{FU(D \rightarrow FD)} (py)$. As the right-hand side inserts a θ , the left-hand side must also be of the form θ . But this means that the restriction relates x of the form $\theta(\dots)$ to y of the form $\eta(\dots)$. On the other hand, $F\text{Inj}_{\rightarrow}$ only relates θ to θ .

0.3.3 Value and Computation Relations

We now revisit the definition of value and computation relations. Recall that the original definitions mentioned quasi-representability. We begin by restating the definitions in terms of representability in the new

category:

Definition 0.6. Let A and A' be value objects. A **value relation** $c : A \multimap A'$ is a relation between the underlying predomains such that:

- c satisfies push-pull.
- c is left-representable.
- Fc is right-representable.

A **computation relation** $d : B \multimap B'$ is defined analogously: an error domain relation satisfying push-pull, such that d is right-representable and Ud is left-representable.

Remark 3. The above definition is equivalent to the usual definition of embedding-projection pair.

Specifically, suppose we are given an embedding $e : A \rightarrow A'$ and a projection $p : FA' \rightarrow FA$ satisfying $\text{id}_{FA} \sqsubseteq p \circ Fe$ and $Fe \circ p \sqsubseteq \text{id}_{FA'}$. Then we can recover a horizontal morphism in the above sense. In particular, let c be the relation $e^*r(A')$ and let d be $r(FA)^*p$. By Lemma ??, c and d satisfy the push-pull property. It is also true that c is left-representable (by e) and d is right-representable (by p).

We now show Fc is order-equivalent to d as follows:

$$\begin{array}{ccccc}
 FA & \xrightarrow{r(FA)} & FA & \xrightarrow{Fc} & FA' \\
 \downarrow \text{id} & & \downarrow Fe & \text{FUL}_c & \downarrow \text{id} \\
 & & FA' & \xrightarrow{r(FA')} & FA' \\
 & & \downarrow p & \text{DNL}_d & \downarrow \text{id} \\
 FA & \xrightarrow{r(FA)} & FA & \xrightarrow{d} & FA'
 \end{array}$$

Now since d is right-representable, Fc is also right-representable.

0.4 Functorial Actions

0.5 Monadic Bind

0.6 Summary of the Model

Values

- Objects: Value objects (A, M_A, i_A) where A is a predomain, M_A is a commutative monoid, and $i_A : M_A \rightarrow \text{Endo}_{\approx}(A)$ is a homomorphism of monoids.
- Vertical morphisms: Morphisms of the underlying predomains that have an action on perturbations.
- Horizontal morphisms: Relations c between the underlying predomains satisfying push-pull, such that c is left-representable and Fc is right-representable.
- Squares: Perturbed squares $f \sqsubseteq_{\text{ptb}_{c_o}^{c_i}} g$

Computations

- Objects: Computation objects (B, M_B, i_B)
- Vertical morphisms: Morphisms of the underlying error domains that have an action on perturbations
- Horizontal morphisms: Relations d between the underlying error domains satisfying push-pull, such that d is right-representable and Ud is left-representable
- Squares: Perturbed squares

0.7 Revisiting the Graduality Proof

Having described the double category in which the casts live, we now return to the graduality proof. We need to verify that the morphisms of predomains/error domains used to define the cast morphisms do in fact extend to perturbations.

We also need to check that the type constructors act on squares.

0.7.1 The Model Construction

1. If c and c' are value relations, then there is a value relation $c \odot c'$. Likewise for computation relations.
2. If c is a value relation, then there is a computation relation Fc .
3. If d is a computation relation, then there is a value relation Ud .
4. If c_1 and c_2 are value relations, then there is a value relation $c_1 \times c_2$.
5. If c is a value relation and d is a computation relation, then there is a computation relation $c \rightarrow d$.

A Omitted Examples and Proofs

We begin with an example showing a concrete pair of type precision c and c' for which the projection for the composition cc' inserts an additional perturbation.

Let $A_1 = (? \multimap ?) \multimap ?$ and $A_2 = ? \multimap ?$ and $A_3 = ?$, and let $c = \text{Inj}_{\multimap} \multimap r(?) : A_1 \sqsubseteq A_2$ and $c' = \text{Inj}_{\multimap}$.

First note that in the semantics, $F(cc')$ and $FcFc'$ are not equal as relations, but they are quasi-equivalent since they are both quasi-left-represented by the same embedding $Fe_{c'} \circ Fe_c$ (see Lemma D.11 in the appendix of the POPL 2025 paper for details). In particular, there are syntactic perturbations $\delta^l \in M_{FA_1}$ and $\delta^r \in M_{FA_3}$ and a square $\alpha : i(\delta^l) \sqsubseteq_{FcFc'}^{Fcc'} i(\delta^r)$.

Consider the DNR square for $F(cc')$, which says that $i(\delta_1) \sqsubseteq_{r(FA_1)}^{F(cc')} p_{F(cc')}$ for some perturbation $\delta_1 \in M_{FA_1}$. As in the proof of Lemma D.13, to define the DNR square for $F(cc')$, we paste the square α on top of the DNR square for $FcFc'$.

As in the proof, we then observe that we must define the projection morphism $p_{F(cc')}$ to be the composition of $i(\delta^r)$ with the projection $p_{FcFc'}$.

The perturbations δ^l and δ^r are obtained as in the proof of Lemma D.7 in the appendix; we can make that construction concrete since we know what d and d' are. In particular, let us focus on δ^r since that is the one used in the definition of the projection morphism for $F(cc')$. Unfolding the proof of Lemma D.7, we see that the square α is obtained as the following horizontal composition:

$$\begin{array}{ccccc}
 FA_1 & \xrightarrow{r(FA_1)} & FA_1 & \xrightarrow{F(cc')} & FA_3 \\
 i(\delta_{FcFc'}^{l,e}) \downarrow & & \downarrow e & & \downarrow i(\delta_{F(cc')}^{r,e}) \\
 FA_1 & \xrightarrow{FcFc'} & FA_3 & \xrightarrow{r(FA_3)} & FA_3
 \end{array}$$

Here, e is the common embedding $Fe_{c'} \circ Fe_c$.

Thus, the perturbation we are interested in, δ^r , is $\delta_{F(cc')}^{r,e}$. Recalling the action of the functor F on quasi-representable relations, $\delta_{F(cc')}^{r,e}$ is simply the (injection into the coproduct of the) perturbation $\delta_{cc'}^{r,e}$. Now recalling the definition of composition of quasi-representable relations, we have $\delta_{cc'}^{r,e} = \delta_{c'}^{r,e} \cdot \text{push}_{c'}(\delta_c^{r,e})$.

Now we can substitute the concrete values of c and c' . We have $c = U(\text{Inj}_{\multimap} \multimap r(FD)) : U(U(D \rightarrow FD) \rightarrow FD) \multimap U(D \rightarrow FD)$ and $c' = \text{Inj}_{\multimap} : U(D \rightarrow FD) \multimap D$.

The perturbation $\delta_{\text{Inj} \rightarrow}^{r,e}$ is the identity since the corresponding embedding doesn't delay, but the perturbation $\delta_c^{r,e}$ is nontrivial. It corresponds to the fact that in the embedding for c , we project in the domain according to $F\text{Inj} \rightarrow$. This perturbation is an actual delay in the case that the dynamic value is a function.

Thus, we have isolated the place in the projection for $F(cc')$ where an additional step may be taken in the relational semantics as compared to the original term semantics.

A.1 Vertical and Horizontal Composition of Squares

Lemma A.1. *Suppose $f_1 \sqsubseteq \text{ptb}_{c_2}^{c_1} g_1$ and $f_2 \sqsubseteq \text{ptb}_{c_3}^{c_2} g_2$. Then $(f_2 \circ f_1) \sqsubseteq \text{ptb}_{c_3}^{c_1} (g_2 \circ g_1)$.*

Proof. Let m_{f_1} and m_{g_1} be the perturbations in the first square, and let m_{f_2} and m_{g_2} be the perturbations in the second square. We compose the two given squares vertically to obtain

$$\begin{array}{ccc}
A_1 & \xrightarrow{c_1} & A'_1 \\
f_1 \downarrow & & \downarrow g_1 \\
A_2 & & A'_2 \\
i_{A_2}(m_{f_1}) \downarrow & & \downarrow i_{A'_2}(m_{g_1}) \\
A_2 & \xrightarrow{c_2} & A'_2 \\
f_2 \downarrow & & \downarrow g_2 \\
A_3 & & A'_3 \\
i_{A_3}(m_{f_2}) \downarrow & & \downarrow i_{A'_3}(m_{g_2}) \\
A_3 & \xrightarrow{c_3} & A'_3
\end{array}$$

Now we commute the inner perturbation with the inner function on both sides. That is, we have $f_2 \circ i_{A_2}(m_{f_1}) = i_{A_3}(f_2^{\rightarrow}(m_{f_1})) \circ f_2$, and likewise $g_2 \circ i_{A'_2}(m_{g_1}) = i_{A'_3}(g_2^{\rightarrow}(m_{g_1})) \circ g_2$, where $f_2^{\rightarrow} : M_{A_2} \rightarrow M_{A_3}$ is the homomorphism of monoids corresponding to f_2 and similarly for $g_2^{\rightarrow}$. Doing so, we see that the above square is equivalent to

$$\begin{array}{ccc}
A_1 & \xrightarrow{c_1} & A'_1 \\
f_1 \downarrow & & \downarrow g_1 \\
A_2 & & A'_2 \\
f_2 \downarrow & & \downarrow g_2 \\
A_2 & & A'_2 \\
i_{A_3}(f_2^{\rightarrow}(m_{f_1})) \downarrow & & \downarrow i_{A'_3}(g_2^{\rightarrow}(m_{g_1})) \\
A_3 & & A'_3 \\
i_{A_3}(m_{f_2}) \downarrow & & \downarrow i_{A'_3}(m_{g_2}) \\
A_3 & \xrightarrow{c_3} & A'_3
\end{array}$$

Thus, we can take the perturbation corresponding to $f_2 \circ f_1$ to be $m_{f_2} \cdot f_2^{\rightarrow}(m_{f_1})$ and the perturbation corresponding to $g_2 \circ g_1$ to be $m_{g_2} \cdot g_2^{\rightarrow}(m_{g_1})$. □

Lemma A.2. *If $f \sqsubseteq \text{ptb}_{c_o}^{c_i} g$ and $g \sqsubseteq \text{ptb}_{c_o'}^{c_i'} h$, then $f \sqsubseteq \text{ptb}_{c_o \odot c_o'}^{c_i \odot c_i'} h$.*

Proof. □

A.2 Lemmas about Push-Pull, Representability

Lemma A.3. *Let $f : A \rightarrow A'$ be a perturbation-preserving morphism. Then the relation $f^*r(A')$ has the push-pull property.*

*Similarly, let $g : A' \rightarrow A$ be a perturbation-preserving morphism. Then the relation $r(A)^*g$ has the push-pull property.*

Proof. First we show how to derive push. The push homomorphism is simply given by $h^\rightarrow$; it remains to show the existence of the following square:

$$\begin{array}{ccc} A & \xrightarrow{f^*r(A')} & A' \\ i_A(m) \downarrow & & \downarrow i_{A'}(h^\rightarrow(m)) \\ A & \xrightarrow{f^*r(A')} & A' \end{array}$$

By definition, it will suffice to construct a square

$$\begin{array}{ccc} A & \xrightarrow{f^*r(A')} & A' \\ i_A(m) \downarrow & & \downarrow i_{A'}(h^\rightarrow(m)) \\ A & & A' \\ f \downarrow & & \downarrow \text{id} \\ A' & \xrightarrow{r(A')} & A' \end{array}$$

Now since f preserves perturbations, the left-hand side of the square is equal to $i_{A'}(h^\rightarrow(m)) \circ f$, so the above is the same as

$$\begin{array}{ccc} A & \xrightarrow{f^*r(A')} & A' \\ f \downarrow & & \downarrow \text{id} \\ A' & \xrightarrow{r(A')} & A' \\ i_{A'}(h^\rightarrow(m)) \downarrow & & \downarrow i_{A'}(h^\rightarrow(m)) \\ A' & \xrightarrow{r(A')} & A' \end{array}$$

The top square is the definition of the restriction of a relation along a morphism, and the second square is a horizontal identity square. Composing these vertically, we are finished.

The pull property is derived in an analogous manner. □

Lemma A.4. *Let $c : A \multimap A'$ be a predomain relation satisfying push-pull. If c is left-representable by e_c , then there is a perturbed square $\text{id} \sqsubseteq \text{ptb}_{e_c^*r(A')}^c \text{id}$ and a perturbed square $\text{id} \sqsubseteq \text{ptb}_c^{e_c^*r(A')} \text{id}$.*

The analogous result is true for any error domain relation $d : B \multimap B'$ satisfying push-pull.

Proof. We first show that $e_c^*r(A')$ is left-representable by e_c . The square $\text{UPL} : e_c \sqsubseteq_{r(A')}^{e_c^*r(A')} \text{id}$ exists by definition of the restriction. The square $\text{UPR} : \text{id} \sqsubseteq_{e_c^*r(A')}^{r(A)} e_c$ is equivalent to the square $e_c \sqsubseteq_{r(A')}^{r(A)} e_c$ which is just an identity square.

Then since c is also left-representable by e_c , we can combine the squares horizontally to obtain the desired squares:

$$\begin{array}{ccccc}
A & \xrightarrow{r(A)} & A & \xrightarrow{c} & A' \\
\text{id} \downarrow & (1) & \downarrow e_c & (2) & \downarrow \text{id} \\
A & \xrightarrow{e_c^* r(A')} & A' & \xrightarrow{r(A')} & A'
\end{array}$$

Here, (1) is the UPR square for $e_c^* r(A')$ while (2) is the UPL square for c . The other square is constructed symmetrically.

The proof for error domain relations is analogous. □

Lemma A.5. *Let $c : A \multimap A'$ be a predomain relation satisfying push-pull. If c is right-representable by p_c , then there is a perturbed square $\text{id} \sqsubseteq \text{ptb}_{r(A)^* p_c}^c \text{id}$ and a perturbed square $\text{id} \sqsubseteq \text{ptb}_c^{r(A)^* p_c} \text{id}$.*

The analogous result is true for any error domain relation $d : B \multimap B'$ satisfying push-pull.

Proof. Analogous to the proof of the previous lemma. □